

Why Rotating Fluids Make Polygons

Abstract

Saturn has a hexagon. Jupiter has an octagon and a pentagon. We explain why polygonal vortex structures arise in two-dimensional rotating fluids through a two-scale mechanism. At the *macro* scale, the Kraichnan inverse cascade and Onsager negative-temperature statistics drive the formation of coherent vortex clusters—established 2D turbulence [Onsager, 1949, Kraichnan, 1967, Bouchet and Venaille, 2012]. At the *meso* scale, we prove that the regular N -gon is a constrained energy minimum for point vortices on a ring (stable for $N \leq 7$ without central vortex, or any N with sufficient central vortex strength), so the polygon arrangement is selected by dynamical stability.

The key is two-dimensionality: the Green’s function of the 2D Laplacian, $G(r) = -\ln r/(2\pi)$, is the unique interaction satisfying $h'(r) < 0$ and $(rh'(r))' = 0$. We prove analytically that any interaction with $h'(r) < 0$ and $(rh'(r))' \leq 0$ makes the N -gon an energy extremum along the spiral deformation direction. This would not hold in 3D, where the Green’s function $-1/(4\pi r)$ has $h' > 0$.

Three planetary configurations are examined: Saturn’s hexagon ($N = 6$), Jupiter’s north polar octagon ($N = 8$), and south polar pentagon ($N = 5$). The mathematical results (energy minimum, sign rule) are proven for discrete point vortices and apply directly to Jupiter’s cyclone rings. Saturn’s hexagon is a continuous jet meander, for which the point vortex model is not directly applicable; instead, the wavenumber $N = 6$ is selected by Rossby wave stationarity of the continuous QGPV equation. The shared logarithmic Green’s function provides a structural link between the two mechanisms, but a rigorous bridge from continuous to discrete has not been established. *Which* polygon depends on planetary parameters; the results are consistent with the hypothesis that polygon formation is a general consequence of two-dimensionality.

1 Introduction

In 1988, David Godfrey examined Voyager images of Saturn’s north pole and found a hexagon [Godfrey, 1988]—a persistent, regular, six-sided polygon inscribed in the jet stream at 76° N, stable across decades. Thirty years later, the Juno spacecraft revealed that Jupiter’s poles harbor their own polygonal arrangements: eight cyclones in an octagonal ring at the north pole, five in a pentagonal ring at the south [Adriani et al., 2018]. Numerical simulations can reproduce these structures [Morales-Juberías et al., 2011, Gavriel and Kaspi, 2021], and vortex crystals emerge spontaneously in 2D turbulence simulations on polar caps [Siegelman et al., 2022].

These observations raise a question with two parts. The first—*how* does a specific polygon form?—has received considerable attention. The second—*why* should different

physical mechanisms on different planets all produce the same geometric outcome?—has not.

We answer the second question. The mechanism operates at two scales:

1. **Macro scale: cluster formation.** The Kraichnan (1967) inverse energy cascade drives 2D turbulence toward large-scale coherent structures. In the Onsager (1949) framework, this produces a negative-temperature state where same-sign vorticity concentrates into N coherent clusters [Bouchet and Venaille, 2012, Siegelman et al., 2022]. This step explains why discrete vortex clusters exist; it does not determine their spatial arrangement.
2. **Meso scale: polygon arrangement.** Given N vortex clusters on a ring of fixed angular momentum, the regular N -gon *minimizes* the interaction energy on the conserved-quantity surface (Theorem 3). By the energy-Casimir method, this makes the N -gon a nonlinearly stable equilibrium for $N \leq 7$ without a central vortex, or for any N with sufficient central vortex strength. The polygon is selected by dynamical stability, not by statistical mechanics.

The meso-scale proof (Theorem 3) applies to discrete point vortices and is directly relevant to Jupiter’s cyclone rings. Saturn’s hexagon is a continuous jet meander, not a discrete vortex ring; its wavenumber $N = 6$ is selected instead by Rossby wave stationarity of the QGPV equation (Proposition 5). Both mechanisms share the logarithmic Green’s function as a structural foundation, but the mathematical arguments differ.

Which polygon forms ($N = 5, 6, 8$) is determined by planet-specific selection criteria.

2 Setup

2.1 Point vortex dynamics on a 2D surface

On a two-dimensional surface, the stream function ψ satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y. \quad (1)$$

The Green’s function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, satisfying $G'(r) = -1/(2\pi r) < 0$ for all $r > 0$.

For N point vortices with equal circulations κ on the collision-free configuration space $\{|z_j - z_k| > 0 \text{ for all } j \neq k\}$, where H is smooth and $\mathbb{Z}_N$ -invariant, the interaction energy is

$$H = - \sum_{j < k} \ln |z_j - z_k|, \quad (2)$$

where we absorb $\kappa^2/(4\pi)$ into the normalization. By the equivariant critical point theorem, the regular N -gon—a fixed point of $\mathbb{Z}_N$ —is a critical point of H .

Point vortex dynamics conserves three quantities: the energy H , the angular impulse $L = \sum |z_k|^2$, and the linear impulse $\mathbf{P} = \sum z_k$. The physically accessible configuration space is the $(2N - 3)$ -dimensional surface $\{L = NR^2, \mathbf{P} = 0\}$.

2.2 The centered spiral deformation

Consider N vortices on a ring of radius R , deformed along the one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (3)$$

At $\sigma = 0$: the regular N -gon. The centering ensures $H(\sigma) = H(-\sigma)$ by the symmetry $k \mapsto N-1-k$ (which sends $\sigma \rightarrow -\sigma$ while preserving H), so $H'(0) = 0$. The spiral deformation is chosen for analytical tractability: it is the natural one-parameter family in the Möbius-group classification (the loxodromic orbit $z \mapsto \lambda z$ with $\lambda = e^{\sigma/N + 2\pi i/N}$ through the N -gon), and it is the unique $\mathbb{Z}_N$ -equivariant deformation that preserves the ring topology while breaking rotational symmetry to discrete. Other perturbation directions (individual Fourier modes, radial displacements) are analyzed through the full constrained Hessian in Theorem 3.

3 Main Results

3.1 Energy curvature along the spiral

Theorem 1 (Energy curvature at the polygon). *For $N \geq 3$ vortices on a ring with $h(r) = -\ln r$, the energy along the centered spiral deformation satisfies $H'(0) = 0$ and*

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0. \quad (4)$$

Proof. Part (a): $H'(0) = 0$ by the centering symmetry $k \mapsto N-1-k$. Part (b): Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ with $u = \sigma/N$. Direct differentiation gives $F''(0, m) = m^2/[2\sin^2(\pi m/N)]$. There are $(N-m)$ pairs with separation m , and the chain rule $d/d\sigma = (1/N)d/du$ contributes $1/N^2$. This derivation uses the specific structure of $h = -\ln r$: the general formula (5) has two terms, but for the logarithmic interaction $(rh')' = (-1)' = 0$, so the S_m terms (first term in (5)) vanish identically. Only the second term survives, giving (4) as a special case of (5) with $A_m = 0$ and $B_m = h'(d_m)m^2(N-m)/(N^2d_m) = -m^2(N-m)/(N^2d_m^2)$. Substituting $d_m = 2\sin(\pi m/N)$ yields (4). Every factor in the sum is strictly positive, so $H''(0) < 0$. $\square$

Warning (energy maximum vs. minimum). Theorem 1 proves $H''(0) < 0$ along the spiral direction *only*. This does **not** imply the N -gon is an energy maximum on the physically accessible surface. The spiral deformation (3) changes the angular impulse $L = \sum |z_k|^2$ at $O(\sigma^2)$, so it leaves the conserved surface at second order. Theorem 3 below proves that on the conserved surface $\{L = NR^2, \mathbf{P} = 0\}$, the N -gon is a constrained energy *minimum* for $N \leq 7$ —the opposite conclusion. The Lagrange multiplier shift $\mu_L = -(N-1)/4$ overwhelms the negative $H''(0)$ when the perturbation is projected onto the constraint tangent space. The reader should carry the *minimum* (not maximum) as the correct physical picture throughout.

3.2 The sign rule (proven)

Theorem 2 (Sign rule for general interactions). *For $N \geq 3$ vortices on a ring with general pairwise interaction $h(r)$, the energy along the centered spiral satisfies*

$$H''(0) = \sum_{m=1}^{N-1} \left[(d_m^2 h''(d_m) + d_m h'(d_m)) S_m + \frac{h'(d_m) m^2 (N-m)}{N^2 d_m} \right], \quad (5)$$

where $d_m = 2R \sin(\pi m/N)$ and $S_m = \sum_{k=0}^{N-m-1} [(2k+m-(N-1))/(2N)]^2 \geq 0$.

If $h'(r) < 0$ and $(rh'(r))' \leq 0$ for all $r > 0$, then $H''(0) < 0$.

Proof. Step 1: Pairwise distance as a function of σ . For vortices j and k with separation $m = j - k > 0$, write the centered exponent as $(j - (N-1)/2)/N = (j' + k')/2 + m/(2N)$ where $j' = (j - (N-1)/2)/N$ and $k' = (k - (N-1)/2)/N$. Define the mean displacement $s = \sigma(j' + k')/2$ and half-separation $t = \sigma m/(2N)$. Then $z_j(\sigma) = R e^{s+t} e^{i\theta_j}$, $z_k(\sigma) = R e^{s-t} e^{i\theta_k}$ with $\theta_j - \theta_k = 2\pi m/N$, giving

$$D(\sigma) := |z_j - z_k|^2 = R^2 e^{2s} [e^{2t} + e^{-2t} - 2 \cos(2\pi m/N)] = R^2 e^{2s} [2 \cosh(2t) - 2 \cos(2\pi m/N)].$$

At $\sigma = 0$: $s = t = 0$, so $D(0) = 2R^2 [1 - \cos(2\pi m/N)] = 4R^2 \sin^2(\pi m/N) = d_m^2$.

Step 2: First and second derivatives of D . Write $c_s = (j' + k')/2$ and $c_t = m/(2N)$, so $s = \sigma c_s$, $t = \sigma c_t$. Direct differentiation gives

$$D'(0) = 4c_s(1 - \cos \phi), \quad \phi = 2\pi m/N,$$

$$D''(0) = 8c_s^2(1 - \cos \phi) + 8c_t^2.$$

Step 3: Chain rule for $h(\sqrt{D})$. Write $d = \sqrt{D}$. Two applications of the chain rule give

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_{\sigma=0} = \underbrace{\frac{h''(d_m)}{4d_m^2} (D')^2 + \frac{h'(d_m)}{2d_m} \left[D'' - \frac{(D')^2}{2d_m^2} \right]}_{\text{evaluated at each pair } (j,k)}.$$

Substituting $D' = 4c_s d_m^2/2$ (since $1 - \cos \phi = d_m^2/(2R^2)$, and absorbing R into d_m) and collecting terms:

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_0 = [d_m^2 h''(d_m) + d_m h'(d_m)] c_s^2 + \frac{h'(d_m)}{d_m} c_t^2 \frac{d_m^2}{(1 - \cos \phi)}.$$

Step 4: Summation over pairs. For fixed separation m , sum over the $N - m$ pairs $(k, k+m)$ for $k = 0, \dots, N-m-1$. Define

$$S_m = \sum_{k=0}^{N-m-1} c_{s,k}^2, \quad \text{where } c_{s,k} = \frac{2k+m-(N-1)}{2N}.$$

Since $\sum_k c_{s,k} = 0$ by the centering, $S_m \geq 0$ with equality only if all $c_{s,k} = 0$ (which requires $N-m=1$, i.e. $m=N-1$). The second sum contributes $c_t^2 \cdot (N-m) = m^2(N-m)/(4N^2)$, and $d_m^2/(1 - \cos \phi) = 4R^2$. Summing over $m = 1, \dots, N-1$ yields (5).

Step 5: Sign determination. Under $h'(r) < 0$: the second term $h'(d_m) m^2(N-m)/(N^2 d_m) < 0$ for all $m = 1, \dots, N-1$ (including $m=N-1$, where $S_{N-1} = 0$ but the second term is still negative). Under $(rh'(r))' \leq 0$: the first coefficient $d_m^2 h''(d_m) + d_m h'(d_m) = d_m \cdot (rh')'|_{r=d_m} \leq 0$, so the first term is ≤ 0 since $S_m \geq 0$. (At $m=N-1$: $S_{N-1} = 0$ and the first term vanishes regardless of the sign condition; the inequality $H''(0) < 0$ follows from the strictly negative second terms alone.) Therefore $H''(0) < 0$. $\square$

Remark 1 (Borderline status of the logarithmic interaction). *The sufficient conditions of Theorem 2 partition interactions into three classes:*

- (i) $h' < 0$ and $(rh')' < 0$: both terms in (5) are strictly negative. Examples: $h = -r^\alpha$ ($\alpha > 0$), for which $(rh')' = -\alpha^2 r^{\alpha-1} < 0$.
- (ii) $h' < 0$ and $(rh')' = 0$: the first term vanishes and only the second contributes. For smooth h on $(0, \infty)$, this class contains a unique interaction: $(rh')' = 0 \Rightarrow rh' = c \Rightarrow h' = c/r \Rightarrow h = c \ln r + b$; combined with $h' < 0$ gives $c < 0$, i.e. $h(r) = -a \ln r + b$ with $a > 0$.
- (iii) $h' < 0$ and $(rh')' > 0$ at some r : the sign of the first term is indefinite and $H''(0) < 0$ is not guaranteed.

The 2D Green's function $h(r) = -\ln r$ falls in class (ii)—the boundary between guaranteed and non-guaranteed negativity. In this precise sense, the logarithmic interaction produces the weakest $|H''(0)|$ among all interactions in classes (i) and (ii).

3.3 The constrained energy minimum

Theorem 3 (Constrained energy minimum). *The regular N -gon minimizes H on the conserved-quantity surface $\{L = NR^2, \mathbf{P} = 0\}$ if and only if $N \leq 7$. The Lagrangian Hessian*

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I, \quad \mu_L = -\frac{N-1}{4}, \quad (6)$$

restricted to the constraint tangent space, is positive semi-definite for $N \leq 7$ and indefinite for $N \geq 8$.

Proof. The energy-Casimir functional is $\mathcal{H} = H + |\mu_L| L$, whose Hessian is (6). The Hessian $\nabla^2 H$ is a $2N \times 2N$ real symmetric matrix that commutes with the $\mathbb{Z}_N$ action $z_k \mapsto z_{k+1 \pmod N}$ (since H is invariant under cyclic permutation of equal-circulation vortices). By Schur's lemma for real representations of finite abelian groups, $\nabla^2 H$ block-diagonalizes in the discrete Fourier basis into 2×2 blocks, one per mode $m \in \{0, \dots, N-1\}$. For mode m , decompose the perturbation at vortex k into radial (δr_k) and tangential (δt_k) components, with Fourier amplitudes $\delta r_k = a \cos(2\pi mk/N)$, $\delta t_k = b \cos(2\pi mk/N)$. The 2×2 Hessian block for the pair (a, b) involves two sums over separations $p = 1, \dots, N-1$:

$$\begin{aligned} H_{rr}^{(m)} &= \sum_p \left[\frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} + \frac{1 + \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} \right] - 2 \sum_p \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)}, \\ H_{rt}^{(m)} &= \sum_p \frac{\sin(2\pi pm/N)}{4 \sin^2(\pi p/N)}, \\ H_{tt}^{(m)} &= -H_{rr}^{(m)}. \end{aligned}$$

The off-diagonal $H_{rt}^{(m)}$ vanishes identically by the anti-symmetry $\sin(2\pi pm/N) \rightarrow -\sin(2\pi(N-p)m/N)$ under $p \rightarrow N-p$, since $\sin^2(\pi p/N) = \sin^2(\pi(N-p)/N)$ (so the denominator is symmetric while the numerator is anti-symmetric; the sum over the full range $p = 1, \dots, N-1$ cancels in pairs). Therefore the block is diagonal: $\text{diag}(H_{rr}^{(m)}, -H_{rr}^{(m)})$, giving eigenvalues $\pm H_{rr}^{(m)} = \pm \sigma_m$.

The diagonal entry $H_{rr}^{(m)}$ simplifies using the Havelock identity (Appendix A): the self-sum $\sum_p 1/[4 \sin^2(\pi p/N)] = (N-1)/2$ (a special case of $T_0 + (N-1)/2$) and the cross-sum

is $T_m = m(N-m)/2$. Thus

$$\sigma_m = H_{rr}^{(m)} = \frac{N-1}{2} - \frac{m(N-m)}{2}, \quad (7)$$

using

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2} \quad (8)$$

(proven in Appendix A; cf. Havelock [1931], §3). The Lagrangian eigenvalues are thus

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}.$$

Since $\lambda_m^- \geq 0$ always, positive semi-definiteness requires $\lambda_m^+ \geq 0$ for the modes tangent to the constraint surface. Three modes are removed: *Mode* $m = 0$ corresponds to uniform dilation $\delta z_k \propto z_k$, which changes $L = \sum |z_k|^2$ and is therefore not tangent to $\{L = NR^2\}$. *Modes* $m = 1$ and $m = N-1$ generate infinitesimal translation ($\delta z_k = \text{const}$, changing $\mathbf{P} = \sum z_k$) and overall rotation ($\delta z_k = iz_k$, a Goldstone mode of the $SO(2)$ symmetry of H). Both have $\sigma_m = 0$, consistent with the continuous symmetry group; they lie in the kernel of $\nabla^2 H$ and produce the zero eigenvalues. The remaining modes $m = 2, \dots, N-2$ are tangent to the constraint surface and carry the stability information. The minimum of λ_m^+ over m occurs at $m = \lfloor N/2 \rfloor$:

$$\min_m \lambda_m^+ = \begin{cases} (N-1)(7-N)/8 & N \text{ odd,} \\ (N-1) - N^2/8 & N \text{ even.} \end{cases}$$

Both expressions are > 0 for $N \leq 6$, equal to zero at $N = 7$, and < 0 for $N \geq 8$. Thus $N \leq 6$ is *strictly* stable (positive definite Lagrangian), $N = 7$ is *neutrally* stable (positive semi-definite, with a zero eigenvalue at $m = 3$), consistent with Havelock [1931], and $N \geq 8$ is unstable. (The $N = 7$ marginal case is further analyzed in Section 6.3.) $\square$

The value $\mu_L = -(N-1)/4$ follows from the constrained stationarity condition $\partial H / \partial x_0 = \mu_L \partial L / \partial x_0$ at vortex 0 (position $(R, 0)$): the left side evaluates to $-\sum_{k=1}^{N-1} (1 - \cos(2\pi k/N)) / [2(1 - \cos(2\pi k/N))] = -(N-1)/2$, and the right side is $\mu_L \cdot 2R$, giving $\mu_L = -(N-1)/(4R^2)$ (equivalently $-(N-1)/4$ for $R = 1$). The equal-contribution property—that each pair $(0, k)$ contributes identically $-1/(2R)$ to $\partial H / \partial x_0$ —follows from $h'(r)/r = -1/r^2$ being independent of r after cancellation with the chord geometry: $(1 - \cos \theta_k) / [4 \sin^2(\theta_k/2)] = 1/2$ for all k . This is a consequence of the *scale invariance* of the logarithmic interaction, and is the unique property that makes $C_1 = N-1$ (see Section 6.7) rather than a k -dependent sum. This shifts all eigenvalues of $\nabla^2 H$ upward by $(N-1)/2$. This overwhelms the $H''(0) < 0$ result along the spiral: the spiral direction, when projected onto the constraint tangent space, has *positive* Lagrangian curvature for all N . The negative curvature of H along the spiral is an artifact of that direction changing the conserved quantity L .

3.4 Central vortex stabilization

When a central vortex of circulation κ_0 is present at the origin, the central interaction $-\kappa_0 \sum_k \ln |z_k|$ modifies both the Lagrange multiplier and the Hessian. Two effects contribute to the critical eigenvalue:

Lagrange multiplier shift. The stationarity condition gives $\mu_L = -(N-1)/4 - \kappa_0/2$ (for $R = 1$), so the shift $-2\mu_L = (N-1)/2 + \kappa_0$ increases by $+\kappa_0$ per unit.

Hessian projection. The central-vortex Hessian at vortex k is the traceless 2×2 matrix $(\kappa_0/R^2) \begin{bmatrix} \cos 2\theta_k & \sin 2\theta_k \\ \sin 2\theta_k & -\cos 2\theta_k \end{bmatrix}$, which has eigenvalue $+\kappa_0/R^2$ for radial perturbations $(\cos \theta_k, \sin \theta_k)$ and $-\kappa_0/R^2$ for tangential. For $N \geq 8$, the critical eigenvector of the ring-only Lagrangian (the most negative eigenvalue direction) is *purely radial* at every vortex position. This follows analytically from the vanishing of the off-diagonal Hessian block $H_{rt}^{(m)}$ (proved in the Theorem 3 derivation above): the mixed sum $\sum_{p=1}^{N-1} \sin(2\pi pm/N)/[4 \sin^2(\pi p/N)] = 0$ vanishes for all m because $\sin(2\pi pm/N)$ is anti-symmetric under $p \rightarrow N-p$ while $\sin^2(\pi p/N)$ is symmetric, so the summand cancels in pairs $(p, N-p)$ over the full range. This decoupling is specific to the logarithmic (harmonic) interaction, for which the radial and tangential Hessian components are equal and opposite ($H_{tt} = -H_{rr}$). Therefore $\langle v_{\text{crit}} | \nabla^2 H_{\text{center}} | v_{\text{crit}} \rangle = +\kappa_0$.

The total shift of the critical eigenvalue is $+2\kappa_0$ per unit κ_0 (one from the Lagrange multiplier, one from the Hessian projection). Setting the shifted eigenvalue to zero:

Proposition 4 (Critical central vortex strength). *For $N \geq 8$, the N -gon with central vortex is a constrained energy minimum if and only if $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$, where*

$$\kappa_{\text{crit}}(N) = \begin{cases} (N^2 - 8N + 8)/16 & N \text{ even}, \\ (N-1)(N-7)/16 & N \text{ odd}. \end{cases} \quad (9)$$

Proof. By Theorem 3, the minimum constrained Lagrangian eigenvalue at $\kappa_0 = 0$ is $\lambda_{\text{min}} = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$. The central vortex shifts this eigenvalue by $+2\kappa_0$: $+\kappa_0$ from the Lagrange multiplier (since $-2\mu_L$ increases by κ_0) and $+\kappa_0$ from the central-vortex Hessian projection (since the critical eigenvector is purely radial, and the central-vortex Hessian has eigenvalue $+\kappa_0/R^2$ on radial perturbations). Setting $(N-1) - m(N-m)/2 + 2\kappa_0 = 0$ with $m = \lfloor N/2 \rfloor$ gives (9). $\square$

The first values are:

N	≤ 7	8	9	10	11	12	13	14
κ_{crit}	0	$\frac{1}{2}$	1	$\frac{7}{4}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{9}{2}$	$\frac{23}{4}$

These agree with numerical bisection on the constrained Lagrangian eigenvalues to within 0.01 for all $N \leq 15$ (supplementary code), and with the classical Thomson dynamical stability analysis [Thomson, 1883, Havelock, 1931], confirming that energy and dynamical stability criteria are equivalent.

For $N = 6$ with central vortex κ_0 , an exact result is available. The 12×12 stability matrix (whose explicit form follows from the Hamiltonian point vortex equations linearized about the hexagonal equilibrium; see Havelock [1931] for the general construction) has characteristic polynomial

$$p(\lambda) \propto \lambda^2 (9 + 36\kappa_0/\kappa + 16\pi^2 \lambda^2) \cdot Q(\lambda, \kappa_0)^2,$$

computed by symbolic algebra (Mathematica; code and intermediate factorization steps in supplementary material). The quadratic factor $9 + 36\kappa_0/\kappa + 16\pi^2 \lambda^2$ vanishes at $\lambda = 0$ when $9 + 36\kappa_0/\kappa = 0$, i.e. $\kappa_0/\kappa = -9/36 = -1/4$. This is the exact critical ratio for the $N = 6$ ring: for $\kappa_0/\kappa > -1/4$ the hexagonal configuration is stable; for $\kappa_0/\kappa < -1/4$ an anticyclonic central vortex destabilizes it.

The center-vs-ring geometry is essential: multi-ring configurations are generically less stable (Section 6.4).

3.5 The two-scale mechanism

The physical picture is:

Scale	Question	Mechanism
Macro	Why do N coherent vortices form?	Onsager + inverse cascade
Meso	Why regular polygon?	Arnold/Thomson stability

The Onsager argument [Onsager, 1949] operates at the macro scale: the inverse cascade and negative-temperature statistics explain why same-sign vorticity concentrates into N coherent clusters, as demonstrated in numerical simulations [Siegelman et al., 2022]. This is a physical hypothesis for planetary atmospheres, not a derived result. Whether the specific planetary configurations (Jupiter’s polar cyclones, Saturn’s hexagonal jet) are actually in or near an Onsager negative-temperature equilibrium—rather than being dynamically maintained metastable states sustained by ongoing convection—is an open empirical question. We assume an effective negative-temperature state is reached through enstrophy dissipation at small scales, but this assumption is not tested here.

The polygon arrangement is then determined by the meso-scale energy landscape: the regular N -gon minimizes the interaction energy at constant angular momentum (Theorem 3). This is a deterministic stability result, not a statistical one.

3.6 Complementary wavenumber selection

Proposition 5 (Complementary selection). *The observed polygon wavenumber N is determined by two complementary mechanisms, both arising from the logarithmic Green’s function:*

1. **Rossby stationarity** (for jet meanders): the QGPV eigenvalue problem selects the wavenumber n^* of the stationary mode ($\omega = 0$). For Saturn: $n^* \approx 5.6 \rightarrow 6$.
2. **Thomson stability** (for discrete cyclone rings): the N -gon is stable for $N \leq 7$ without central vortex, or for N satisfying $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ with one.

For Saturn, n^* is determined by the barotropic Rossby dispersion relation: a stationary wave on a zonal jet of speed U satisfies $\omega = Uk - \beta k/(k^2 + l^2) = 0$, giving the stationary speed $U^* = \beta/(k^2 + l^2)$ [Pedlosky, 1987, Vallis, 2017]. For Saturn’s hexagonal jet at 76°N : $\beta = 2\Omega \cos \varphi/R_S \approx 1.46 \times 10^{-12} \text{ m}^{-1} \text{ s}^{-1}$ (with $\Omega = 1.64 \times 10^{-4} \text{ rad/s}$, $R_S = 5.4 \times 10^7 \text{ m}$), and the Gaussian jet half-width $\sigma_{\text{jet}} \approx 2.5 \times 10^6 \text{ m}$ sets $l \sim 1/\sigma_{\text{jet}}$. This gives $n^* \approx 5.6$, rounding to $n = 6$ [Sánchez-Lavega et al., 2014]. Thomson allows $N \leq 7$; both criteria are satisfied at $N = 6$.

For Jupiter’s north pole ($N = 8$): the cyclones are discrete vortex clusters, not a jet meander; Rossby stationarity gives $n^* \approx 0.3$ (not physically relevant). Thomson requires $\kappa_0/\kappa \geq 0.50$, provided by the observed central polar cyclone [Adriani et al., 2018]. For Jupiter’s south pole ($N = 5$): stable without central vortex.

3.7 Eigenvalue branch disconnection

Within the WKB approximation, the Rayleigh–Kuo eigenvalue equation $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$ has real potential V , so the exponent satisfies $s^2 \in \mathbb{R}$. Writing $s = \sigma + i\alpha_\rho$: $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$. When $V < 0$ (verified for Saturn: $V = -364$), $\alpha_\rho \neq 0$ forces $\sigma = 0$. The elliptic

($\sigma = 0$) and hyperbolic ($\alpha_\rho = 0$) branches are disconnected: smooth jet perturbations cannot destroy the polygon at leading WKB order.

4 Planetary Verification

4.1 Measuring polygon quality: σ_{geom}

For N features at positions z_k , define

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N |\ln |z_k/z_k^{\text{ideal}}||, \quad (10)$$

where z_k^{ideal} are the best-fit regular N -gon vertices.

4.2 Cross-planetary comparison

Table 1: Planetary polygon configurations. κ_0/κ is the ratio of central to ring cyclone circulation. For Jupiter north, this is estimated from Juno imagery [Adriani et al., 2018] using the cyclone area ratio as a proxy; assuming approximately uniform vorticity within each cyclone, $\kappa \propto \omega \cdot A$, so the circulation ratio equals the area ratio times the vorticity ratio. Both cyclone types have comparable peak vorticities, giving $\kappa_0/\kappa \approx A_0/A_{\text{ring}} \sim 1$ with a systematic uncertainty of order 30% from the vorticity profile assumption. This does not affect the qualitative conclusion ($\kappa_{\text{crit}} = 0.50$ for $N = 8$). “—” indicates the value is not relevant (since $N = 5 \leq 7$, no central vortex is required).

System	N	σ_{geom}	κ_0/κ	Selection mechanism
Saturn hexagon	6	0.50	N/A (jet)	Rossby stationarity
Jupiter north	8	0.06	~ 1	Thomson + central cyclone
Jupiter south	5	0.05	—	Thomson (intrinsically stable)

A direct comparison of σ_{geom} across systems requires caution. Saturn’s value (0.50) is an order of magnitude larger than Jupiter’s (0.05–0.06) because the two systems are physically different: Saturn’s hexagon is a continuous jet meander with amplitude $\varepsilon \approx 0.11 \pm 0.11$ (100% uncertainty from the Gaussian jet profile assumption—this cannot serve as quantitative verification), while Jupiter’s cyclone rings are discrete vortices closely approximating a regular polygon. The σ_{geom} metric measures geometric deviation from a perfect polygon, but its physical meaning differs between a jet meander (where σ reflects the wave amplitude) and a vortex ring (where σ reflects positional scatter). We present these as two physically distinct realizations that both produce polygonal geometry, not as instances of a single quantitative prediction.

4.3 Broader observational context

Beyond the three planetary configurations, the theoretical stability boundary $\kappa_{\text{crit}}(N)$ from Proposition 4 can be compared with laboratory and numerical observations:

Laboratory experiments. Jansson et al. [2006] and Barbosa Aguiar et al. [2010] observe polygons with $N = 2$ through 6 in rotating-tank experiments by varying the differential

rotation rate. All observed N values fall in the $\kappa_{\text{crit}} = 0$ range ($N \leq 7$), consistent with the Thomson stability boundary. No laboratory experiment has produced a stable polygon with $N \geq 8$ without a central vortex—also consistent. (The laboratory mechanism is barotropic instability of a shear layer, not Thomson point vortex dynamics, so the agreement is structural rather than quantitative.)

Numerical simulations. Siegelman et al. [2022] produce $N = 8$ (north) and $N = 5$ (south) vortex crystals from 2D turbulence on polar caps, with a central polar cyclone emerging alongside the ring in both cases. The central cyclone provides the stabilization required by Proposition 4 for $N = 8$.

Uranus. Akins et al. [2023] detect a single polar cyclone at Uranus’s north pole—a single-vortex ($N = 1$) configuration, below the polygon threshold. No ring structure has been observed. Whether the absence of a Uranian polygon reflects insufficient vortex formation (macro-scale constraint) or insufficient observations (the pole has been poorly resolved) remains open.

Neptune. No polar polygon has been observed. Voyager 2 imagery is insufficient to resolve ring structure.

All observations are consistent with the theoretical stability boundary: $N \leq 7$ is observed without a central vortex, $N = 8$ is observed only with one, and no $N \geq 9$ ring has been reported in any system.

5 Discussion

5.1 Why 2D and not 3D

The argument depends critically on two-dimensionality. The 2D Green’s function $G(r) = -\ln r/(2\pi)$ is the unique interaction satisfying three properties: (a) $h'(r) < 0$ (class (ii) of Remark 1: the sign rule applies); (b) $(rh'(r))' = 0$ (the unique solution is $h = -a \ln r + b$; this is the boundary between classes (i) and (iii)); (c) $\nabla^2 h = 0$ for $r > 0$ (harmonic — h is the Green’s function of the 2D Laplacian). In 3D, the Green’s function $G(r) = -1/(4\pi r)$ has $G'(r) = 1/(4\pi r^2) > 0$ —the sign rule is violated, the N -gon is not an energy extremum along the spiral, and the stability analysis gives different results. The present mechanism—sign rule plus energy-Casimir stability—does not operate in 3D; whether other mechanisms can produce persistent polygonal vortex structures in three dimensions is a separate question. Extensions to curved 2D surfaces and finite deformation radius are developed in Section 6.

5.2 What is novel

The individual components—Onsager condensation, Kraichnan cascades, Thomson stability, 2D Green’s functions—are established. What is new is: (1) the proof that the regular N -gon is a *constrained energy minimum* on the conserved-quantity surface (correcting the common assumption that it is an energy maximum); (2) the analytic sign rule (Theorem 2) proving $H''(0) < 0$ for all interactions with $h' < 0$ and $(rh')' \leq 0$; (3) the two-scale separation showing the Onsager argument applies to cluster *formation* while the polygon *arrangement* is deterministic stability; (4) the complementary selection framework unifying Saturn (Rossby) and Jupiter (Thomson); (5) the observation that the logarithmic interaction’s borderline status in the sign rule makes it the “least robust” polygon producer—any other qualifying interaction gives stronger selection; (6) the

spherical stability boundary $\varphi_{\text{crit}}(N)$ showing the sphere *destabilizes* relative to the plane (the Lagrange shift vanishes at the equator), with Saturn’s hexagon inside the stable zone; (7) the K_0 suppression transition with two distinct phase boundaries (spiral and constrained) and the class (iii) interaction landscape; and (8) the vortex blob bridge showing regularization preserves and enhances stability, resolving the $N = 7$ marginal case, with profile independence via a Fourier dominance argument (Section 6).

Relation to prior work. The connection between Onsager statistics and large-scale vortex formation is developed in Bouchet and Venaille [2012] and Chavanis and Sommeria [1996], who classify self-organized vortex states in bounded domains using statistical mechanical methods. Aref et al. [2003] reviews vortex crystal configurations and their stability. The present contribution differs in three respects: (i) the closed-form formula for $H''(0)$ (equation (4)) does not appear in the Thomson–Havelock stability literature or in these reviews, which analyze stability through eigenvalues of the dynamical matrix rather than energy curvature along a specific deformation; (ii) the analytic sign rule (Theorem 2) identifying the precise conditions on h that guarantee $H''(0) < 0$ is new; and (iii) the constrained Hessian result (Theorem 3), showing the N -gon is an energy *minimum* rather than maximum on the conserved surface, corrects a widespread assumption in the Onsager vortex literature that polygonal configurations are energy maxima selected by negative temperature.

5.3 Limitations and scope

The mathematical results—the constrained energy minimum (Theorem 3), the sign rule (Theorem 2), and the stability table—are proven for discrete point vortices. They apply directly to Jupiter’s polar cyclone rings. For Saturn’s hexagon, which is a continuous jet meander, the relevant mechanism is Rossby wave stationarity of the QGPV equation, not the point vortex energy landscape. The continuous and discrete descriptions share the logarithmic Green’s function; the vortex blob analysis (Section 6.6) establishes that the stability boundary is preserved under regularization and that the $N = 7$ marginal case is resolved. The Saturn bridge (Section 6.6) connects the continuous QGPV eigenvalue problem to the point vortex Hessian through the critical-layer concentration in the sharp-jet limit; a rigorous proof of the critical-layer convergence (showing the cat’s-eye patches satisfy the conditions of Proposition 8) requires nonlinear critical-layer theory beyond the scope of this paper.

The Onsager negative-temperature hypothesis is a physical assumption whose applicability to these specific planetary configurations is not tested here. Cassini and Juno data are digitized from published figures.

Three configurations from two planets is consistent with the proposed mechanism but not sufficient to establish universality. Uranus and Neptune observations would substantially strengthen the case. We use “consistent with a general mechanism” rather than “universal” to reflect the available evidence.

6 Extensions

The results of Sections 3–5 are proven for point vortices on the Euclidean plane with logarithmic interaction. This section extends the analysis to curved surfaces, finite defor-

mation radius, the $N = 7$ marginal case, multi-ring configurations, and the continuous-to-discrete bridge via the vortex blob limit.

6.1 Curved 2D surfaces: the spherical stability boundary

On the 2-sphere, the point vortex interaction $h(\theta) = -\ln(2 \sin(\theta/2))$ satisfies $h' < 0$ and $(\sin \theta \cdot h')' = \frac{1}{2} \sin \theta > 0$ (class (iii)). While $H''(0) > 0$ along the spiral at all tested colatitudes (the polygon is a spiral-direction energy minimum even without the Lagrange shift), the *constrained* Hessian tells a different story.

The angular impulse on the sphere is $L = \sum \cos \theta_k$, with Hessian $\partial^2 L / \partial \theta_k^2 = -\cos \varphi_0$ at colatitude φ_0 . The Lagrangian shift from μ_L is therefore proportional to $\cos \varphi_0$: full near the pole ($\varphi_0 \rightarrow 0$), vanishing at the equator ($\varphi_0 = 90^\circ$). This latitude-dependent shift produces a *colatitude-dependent stability boundary*:

N	3	4	5	6	7	≥ 8
φ_{crit}	$>85^\circ$	54.7°	45.0°	26.6°	$\approx 0^\circ$	0°

The N -gon is a constrained energy minimum on the sphere if and only if $\varphi_0 < \varphi_{\text{crit}}(N)$. The dimensionless eigenvalue $\bar{\lambda} = \sin^2 \varphi_0 \cdot \lambda$ recovers the Havelock values to within 1% at $\varphi_0 = 3^\circ$ ($N=5$: $\bar{\lambda} = 0.99$, Havelock = 1.0; $N=6$: 0.49 vs 0.5; $N=8$: -1.01 vs -1.0).

Saturn's hexagon at 76° N ($\varphi_0 = 14^\circ$) lies inside $\varphi_{\text{crit}}(6) = 26.6^\circ$: the hexagonal ring is in the stable zone. If the hexagon were at 60° N ($\varphi_0 = 30^\circ > 26.6^\circ$), it would cross the boundary and the pattern would not persist. Jupiter's octagon ($N = 8$) is unstable at all colatitudes, consistent with the need for a central vortex.

On the flat torus, $H''(0) < 0$ approaches the planar value for polygons small relative to the domain, with periodic corrections entering at the domain scale.

6.2 Finite Rossby deformation radius

In quasi-geostrophic dynamics with finite Rossby deformation radius R_d , the Green's function is $K_0(r/R_d)$ (modified Bessel function), which recovers $-\ln r$ for $r \ll R_d$. The interaction $h(r) = K_0(r/R_d)$ satisfies $h'(r) = -K_1(r/R_d)/R_d < 0$ and $(rh'(r))' = rK_0(r/R_d)/R_d^2 > 0$ (class (iii)). Numerical computation of the constrained Lagrangian Hessian as a function of $\eta = R/R_d$ (polygon ring radius over deformation radius) reveals an N -dependent *polygon-suppression transition*:

N	Barotropic λ_{min}	Critical R/R_d	Status
≤ 5	≥ 1	none	Stable for all R/R_d
6	$1/2$	≈ 1.29	Suppressed for $R/R_d > 1.29$
7	0	≈ 0	Unstable for any finite R/R_d
≥ 8	< 0	N/A	Unstable (barotropic)

Polygons with $N \leq 5$ are robust against stratification effects. The hexagon ($N = 6$) is the first polygon sensitive to R_d : it requires $R/R_d < 1.29$. The heptagon ($N = 7$), neutrally stable in the barotropic limit, is destabilized by any finite R/R_d —providing an additional mechanism for its rarity beyond the quartic confinement.

For Saturn, the barotropic deformation radius (~ 2000 – 3000 km) gives $R/R_d \approx 8$ – 12 at the hexagonal jet, placing it above the critical ratio. The observed hexagonal persistence

requires the deep barotropic layer to have an effectively large R_d , consistent with Cassini gravity-field measurements of deep zonal winds [Sánchez-Lavega et al., 2014].

The critical ratio is defined implicitly by the equation $\bar{\lambda}(\eta_{\text{crit}}, N) = 0$, where the dimensionless eigenvalue $\bar{\lambda}(\eta, N) = R^2 \cdot \lambda_{\min}(\eta)$ interpolates smoothly from the Havelock value $\bar{\lambda}(0, N) = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$ to zero.

The Lagrange multiplier has the closed form

$$-2\mu_L = \frac{\eta}{R} \sum_{p=1}^{N-1} \frac{(1 - \cos \theta_p) K_1(2\eta \sin(\pi p/N))}{2 \sin(\pi p/N)}, \quad \theta_p = 2\pi p/N, \quad (11)$$

which reduces to $(N-1)/(2R^2)$ as $\eta \rightarrow 0$ (using $K_1(x) \sim 1/x$ and $\eta/R = 1/R_d$). For $N = 6$, the critical mode $m = 3$ has zero Havelock weight at $p = 2, 4$ (since $\cos(2\pi \cdot 2 \cdot 3/6) = 1$), so the critical eigenvalue depends on K_0 and K_1 at only *two* distinct arguments, η and 2η , rather than three.

The K_0 analog of the Havelock identity does not admit a closed form: the summand $K_0(2\eta \sin(\pi p/N))$ is transcendental in $\sin(\pi p/N)$, so the orthogonality trick that closes the logarithmic sum to $m(N-m)/2$ fails. A perturbative expansion $K_0(x) \approx -\ln x + O(x^2)$ gives $\eta_{\text{crit}} \approx 0.9$ for $N = 6$ —30% below the exact value—because the critical ratio lies in the non-perturbative regime where K_0 departs significantly from $-\ln$. The transition occurs precisely where the exponential decay of K_0 (destroying long-range correlations between non-adjacent vortices) overtakes the logarithmic regime, making the suppression inherently non-perturbative. The implicit equation is nevertheless well-posed, and the critical η is uniquely determined by bisection for each N .

The two phase boundaries. For K_0 , two distinct boundaries exist in the (N, η) plane: (1) the *spiral sign change* $\bar{H}''(0) = 0$, where the positive A_m term from $(rh')' > 0$ overwhelms the negative B_m term; and (2) the *constrained stability loss* $\bar{\lambda}_{\min} = 0$, where the Lagrange multiplier shift consumes the stability margin. The constrained boundary ($\eta \approx 1.29$ for $N = 6$) is always *inside* the spiral boundary ($\eta \approx 3.2$ for $N = 6$) because the Lagrange shift reduces the margin before H'' changes sign. Physically, the polygon requires long-range correlations between non-adjacent vortices. When $\eta \gg 1$ and the K_0 interaction decays exponentially, only nearest-neighbor correlations survive and the polygon loses its energetic advantage.

For comparison, the Cauchy-blob regularization $-\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ has $H'' < 0$ for *all* ε —the soft-core never flips the spiral sign. The Gaussian-screened interaction $-\ln(d) \cdot e^{-d^2/L^2}$ shows an earlier transition ($\eta \approx 0.9$ for $N = 6$) due to its faster-than-exponential decay.

The unified stability boundary. The interaction class determines N_{crit} :

Class	Condition	N_{crit}
(i)	$(rh')' < 0$	≥ 7 (increases with $ (rh')' $)
(ii)	$(rh')' = 0$ (only $-\ln r$)	$= 7$ exactly
(iii)	$(rh')' > 0$ (e.g. K_0)	≤ 7 (decreases with screening)

Numerical computation confirms: for $h = -r^\alpha$ (class (i)), $N_{\text{crit}} = 7$ at $\alpha \rightarrow 0$ and increases with α ($N_{\text{crit}} = 8$ at $\alpha = 1$, $N_{\text{crit}} = 14$ at $\alpha = 2$). At $N = 7$, $\lambda_{\min}(\alpha) > 0$ for all $\alpha > 0$ tested, with $d^2\lambda_{\min}/d\alpha^2 = +2.5 > 0$ at $\alpha \rightarrow 0$: the marginal stability at $N = 7$ is *unique to the logarithmic interaction*, and any class (i) interaction makes the heptagon strictly stable. For $K_0(r/R_d)$ (class (iii)), the dimensionless eigenvalue $\bar{\lambda}(\eta)$ decreases

monotonically in η for each Fourier mode (verified for $N = 5, 6$), consistent with the physical expectation that screening weakens the stabilizing long-range correlations. For $N_{\text{crit}} = 6$ at $R/R_d = 1$ and decreases to 5 at $R/R_d = 3$. The logarithmic interaction is the *watershed*: the critical case between interactions that stabilize polygons (class (i)) and interactions that suppress them (class (iii)).

Proposition 6 (Min/max duality). *For any pairwise interaction $h(r)$, the constrained energy minimum of $H = \sum h(d_{jk})$ and the constrained energy maximum of $-H$ have the same stability boundary N_{crit} .*

Proof. The Lagrange multiplier for $-H$ is $\mu_L^{-H} = -\mu_L^H$ (from $-\nabla H = (-\mu_L)\nabla L$). The Lagrangian Hessian $\mathcal{L}_{-H} = -\nabla^2 H + 2\mu_L I = -\mathcal{L}_H$. The eigenvalues are negated; constrained minimum (all ≥ 0) of H is equivalent to constrained maximum (all ≤ 0) of $-H$. $\square$

For gravity ($h = -1/r$, $h' > 0$), the N -gon is a constrained energy maximum. The duality gives $N_{\text{crit}}^{\text{gravity}} = N_{\text{crit}}^{\text{Coulomb}}$. Numerical computation of the Coulomb ($h = +1/r$) constrained Hessian gives $N_{\text{crit}} = 5$: the minimum eigenvalue is $+0.73$ at $N = 5$ and -0.02 at $N = 6$. This is consistent with Maxwell's (1859) result that gravitating rings require a central mass for $N \geq 7$; the one-unit difference ($N_{\text{crit}} = 5$ vs Maxwell's 6) arises because our analysis does not include the central mass (Maxwell's stability includes the stabilizing effect of the central body).

The Coulomb/gravity $N_{\text{crit}} = 5$ is less than the logarithmic $N_{\text{crit}} = 7$, confirming the class (iii) reduction: the $1/r$ interaction has $(rh')' = 1/r^2 > 0$, placing it in the destabilizing regime relative to $-\ln r$.

6.3 The $N = 7$ bifurcation

At $N = 7$, the minimum constrained Lagrangian eigenvalue is zero (mode $m = 3$). To classify the bifurcation, we compute the quartic term of H along the $m = 3$ neutral direction on the constraint surface. The $\mathbb{Z}_7$ -equivariant normal form is $H = (\lambda/2)|z|^2 + (\alpha/4)|z|^4 + \dots$ where z is the complex amplitude of the $m = 3$ mode and $\lambda = 0$ at $N = 7$. Numerical evaluation (tracing H along the Newton-projected constraint surface) gives $\alpha \approx +1.9 > 0$ (a numerical result; analytic derivation of α from the fourth-order energy-Casimir expansion remains open). Since $\alpha > 0$, the energy-Casimir functional $\mathcal{H} = H + |\mu_L|L$ has a strict local minimum along the $m = 3$ direction at quartic order: $\mathcal{H} \geq \mathcal{H}_0 + (\alpha/4)\varepsilon^4$ for $|\varepsilon|$ small. By the Dirichlet stability criterion (a strict local minimum of a conserved functional implies Lyapunov stability; see Boatto and Cabral [2003], Theorem 2.1), the heptagonal configuration is nonlinearly stable. The weaker quartic confinement (compared to the quadratic confinement for $N \leq 6$) explains the relative rarity of heptagonal vortex configurations in observations and simulations.

6.4 Multi-ring configurations

The stabilization in Proposition 4 requires the additional vortex to be at the *center* (the $\mathbb{Z}_N$ fixed point), not merely on an inner ring. Two-ring configurations (M inner vortices at radius R_1 , N outer at R_2) are a classical topic [Aref et al., 2003]; our constrained Hessian computation for representative cases (tested for $M = 1, \dots, 8$, $N = 5, 6, 8$, $R_2/R_1 \in \{1.5, 2, 3\}$; supplementary code)—e.g., $(N, M) = (8, 4)$ at $R_2/R_1 = 2$ gives

3 negative Lagrangian eigenvalues; $(6, 3)$ at $R_2/R_1 = 2$ gives 3 negative eigenvalues—shows that the inter-ring interactions break the single-ring $\mathbb{Z}_N$ symmetry and introduce destabilizing modes absent from the center-plus-ring geometry. This is consistent with Jupiter’s observed configuration.

6.5 BEC vortex predictions

In a quasi-2D Bose–Einstein condensate in a circular trap, quantized vortices interact logarithmically at scales above the healing length ξ . The circulation is quantized: $\kappa = q \cdot h/m$ with integer winding number q . The κ_0/κ ratio of Proposition 4 becomes the winding number ratio q_0/q , yielding sharp integer-valued predictions:

Configuration	κ_{crit}	Prediction
$N \leq 7$ ring	0	Stable without central vortex
$N = 8$ ring	$1/2$	Unstable alone; stable with $q_0 \geq 1$
$N = 9$ ring	1	Marginally stable with $q_0 = 1$; stable with $q_0 \geq 2$
$N = 10$ ring	$7/4$	Unstable with $q_0 = 1$; stable with $q_0 \geq 2$

The $N = 8$ prediction is the sharpest: a ring of 8 unit-winding vortices should be unstable alone and stable upon adding a single central vortex ($q_0 = 1 > \kappa_{\text{crit}} = 1/2$).

The $N = 7$ case offers a distinct signature. The quartic restoring force ($\alpha \approx +1.9$, Section 6.3) produces an *amplitude-dependent* oscillation frequency for the $m = 3$ mode: $\omega \propto \varepsilon$ (where ε is the perturbation amplitude), rather than the constant ω of the quadratic modes at $N \leq 6$. This nonlinear frequency shift is measurable through BEC vortex imaging.

The healing length ξ provides a natural blob width $\varepsilon = \xi/R$. For typical BEC parameters ($\xi \sim 0.3 \mu\text{m}$, $R \sim 50 \mu\text{m}$): $\varepsilon \approx 0.006$, well within the point vortex regime. The blob correction (Proposition 7) is negligible: $\lambda(0.006) \approx \lambda_0 + O(10^{-4})$.

6.6 The continuous-to-discrete bridge

The results of Sections 3–5 are stated for point vortices. Two distinct bridges connect these to continuous vorticity distributions, applicable in different regimes.

6.6.1 The vortex blob bridge (Jupiter regime)

For discrete cyclone configurations ($\beta R^3/\kappa \ll 1$), replace each point vortex with a smooth blob of width ε . The regularized interaction $h_\varepsilon(d) = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ recovers $-\ln d$ as $\varepsilon \rightarrow 0$.

Proposition 7 (Blob stability correction). *The constrained Lagrangian eigenvalues for h_ε satisfy*

$$\lambda_m(\varepsilon) = \lambda_m(0) + \left[-\frac{N^2-1}{12} + P_m\right] \varepsilon^2 + O(\varepsilon^4), \quad (12)$$

where $\lambda_m(0)$ is the Havelock eigenvalue (Theorem 3).

Proof. Lagrange multiplier shift. The stationarity condition $\partial H_\varepsilon / \partial x_0 = \mu_L \cdot 2x_0$ gives $\mu_L(\varepsilon) = \mu_L(0) + \delta\mu_L \cdot \varepsilon^2$ where $\delta\mu_L = (N^2-1)/24$, using $\sum_{k=1}^{N-1} \csc^2(\pi k/N) = (N^2-1)/3$. The shift $-2\delta\mu_L = -(N^2-1)/12$ is *universal* (the same for all Fourier modes).

Hessian correction. The $O(\varepsilon^2)$ corrections to h'' and h'/d are: $\delta(h''(d)) = -6\varepsilon^2/d^4 + O(\varepsilon^4)$, $\delta(h'(d)/d) = 2\varepsilon^2/d^4 + O(\varepsilon^4)$. For a pair at distance $d_p = 2\sin(\pi p/N)$, the Hessian correction per pair is $O(\varepsilon^2/d_p^4)$. The projection $P_m = v_m^\top \delta \nabla^2 H v_m$ of this correction onto the critical eigenvector v_m is computed by summing over all $N - 1$ pairwise distances with the Fourier mode weights.

Operator norm bound. Since $d_{\min} = 2\sin(\pi/N) > 0$: $\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq (2N-3) \cdot 6\varepsilon^2/d_{\min}^4 = O(\varepsilon^2)$. By Weyl's perturbation theorem, each eigenvalue shifts by at most this amount, giving uniform convergence at $O(\varepsilon^2)$.

Profile independence. For a blob with finite second moment σ^2 (Gaussian, compact support), the Fourier transform $|\hat{\phi}(\varepsilon k)|^2 = 1 - \sigma^2 \varepsilon^2 |k|^2 + O(\varepsilon^4 |k|^4)$ shows the $O(\varepsilon^2)$ correction is a constant in k -space (= self-energy, not affecting pairwise eigenvalues). The pairwise correction is $O(\varepsilon^4)$ —*faster* than the Cauchy blob. Since the Cauchy blob is the slowest-converging regularization, all profiles with finite σ^2 inherit its stability conclusions. $\square$

The Hessian projection P_m and the total correction $c_m = -(N^2-1)/12 + P_m$ (specific to the Cauchy blob $h_\varepsilon = -\frac{1}{2}\ln(d^2 + 2\varepsilon^2)$) are exact rational numbers, derived from the finite sum of the $O(\varepsilon^2)$ Hessian correction $\delta H_{xx} = (2\varepsilon^2/d^6)(d_y^2 - 3d_x^2)$ over all pairs, projected onto the critical radial Fourier mode:

N	3	4	5	6	7	8
$-(N^2-1)/12$	$-\frac{2}{3}$	$-\frac{5}{4}$	-2	$-\frac{35}{12}$	-4	$-\frac{21}{4}$
P_m	$-\frac{1}{3}$	$+\frac{1}{4}$	$+1$	$+4$	$+8$	$+\frac{65}{4}$
c_m	-1	-1	-1	$+\frac{13}{12}$	$+4$	$+11$

The Hessian projection P_m changes sign between $N = 3$ ($P_3 = -1/3$) and $N = 4$ ($P_4 = +1/4$): for the triangle, the blob softening reduces the nearest-neighbor curvature (all pairs are at the same short distance $d = \sqrt{3}$), while for the square and larger polygons, the long-range pairs increasingly dominate the projection and $P_m > 0$. Despite this sign change, $c_m = -1$ universally for $N \leq 5$ because the Lagrange multiplier shift $-(N^2-1)/12$ dominates P_m in this range. For $N \geq 6$, P_m grows faster and overwhelms the shift.

For blob profiles with finite second moment σ^2 (Gaussian, compact support), the $O(\varepsilon^2)$ correction is confined to the self-energy and does not affect pairwise eigenvalues (Section 6.6, profile independence argument). The pairwise correction is $O(\varepsilon^4)$ with a profile-dependent coefficient. Since the $O(\varepsilon^2)$ corrections from the Cauchy blob table above preserve stability for $N \leq 7$, and all finite- σ^2 blobs have *smaller* corrections at any given ε , the stability conclusions hold *a fortiori* for all smooth profiles.

Three consequences follow:

(i) *Stability preserved.* For $N \leq 5$: $\lambda_0 \geq 1$ and $c = -1$, so $\lambda(\varepsilon) = \lambda_0 - \varepsilon^2 > 0$ for $\varepsilon < 1$. For $N = 6$: $c > 0$, so $\lambda(\varepsilon) > \lambda_0$ —the blob *enhances* stability.

(ii) *Marginal mode resolved.* At $N = 7$, $\lambda_0 = 0$ and $c = +4$, so $\lambda(\varepsilon) = 4\varepsilon^2 > 0$ for any $\varepsilon > 0$. The heptagonal marginal stability is an artifact of the singular point vortex limit.

(iii) *Instability partially compensated.* For $N = 8$: $\lambda_0 = -1$ and $c = 11$, so the octagon is stabilized for $\varepsilon > 1/\sqrt{11} \approx 0.30$.

6.6.2 Full infinite-dimensional convergence

Proposition 7 addresses center-position perturbations of N smooth blobs. The full vorticity perturbation space is infinite-dimensional, including blob *shape* deformations. A com-

plete stability analysis requires the Hessian of the 2D Euler energy $E[\omega] = \frac{1}{2} \iint \omega(x) G(x, y) \omega(y) dx dy$ at the blob configuration $\omega_\varepsilon = \sum_k \kappa \phi_\varepsilon(\cdot - z_k)$, decomposed into three sectors:

(a) *Center translations.* The Hessian restricted to the $2N$ -dimensional subspace of center displacements is *exactly* the blob Hessian of Proposition 7. Convergence to the point vortex Hessian at $O(\varepsilon^\alpha)$ follows from the convolution estimates.

(b) *Shape perturbations.* Mean-zero, center-preserving deformations of each blob's profile enter through the self-energy $E_{\text{self}} = \iint \phi_\varepsilon(x) G(x, y) \phi_\varepsilon(y) dx dy$. The second variation is the H^{-1} Sobolev inner product on mean-zero functions—positive definite, with smallest eigenvalue scaling as c/ε^2 from the rescaling $\phi_\varepsilon(x) = \varepsilon^{-2} \phi(x/\varepsilon)$. Shape modes are *unconditionally stable*.

(c) *Center-shape coupling.* The cross terms involve overlap integrals $\iint \phi_\varepsilon(x - z_j) \nabla G(x, y)$ [shape per between non-adjacent blobs. For $|z_j - z_k| \gg \varepsilon$, these are $O(\varepsilon^{-1} e^{-d_{\min}/\varepsilon})$ —exponentially small.

Proposition 8 (Inertia preservation). *For sufficiently small $\varepsilon > 0$, the full Hessian of $E[\omega_\varepsilon] + \mu_L L[\omega_\varepsilon]$ on the space of $\mathbb{Z}_N$ -symmetric vorticity perturbations has the same inertia (number of positive, zero, and negative eigenvalues along the constrained directions) as the point vortex Lagrangian Hessian of Theorem 3. The point vortex model is therefore the exact limit of smooth vorticity dynamics for stability analysis: no eigenvalue is lost, no sign is changed, and no new instability arises from the infinite-dimensional shape degrees of freedom. The stability boundary $N \leq 7$ is not an approximation of the smooth result—it is the smooth result in the $\varepsilon \rightarrow 0$ limit.*

Proof. The full Hessian on the constrained perturbation space has block-diagonal structure (up to exponentially small coupling from (c)): the center sector (dimension $2N-3$, eigenvalues converging to Havelock) and the shape sector (infinite-dimensional, all eigenvalues $\geq c/\varepsilon^2 > 0$). Since the shape eigenvalues are positive and bounded away from zero, they cannot produce instabilities. Since the center-shape coupling from (c) is $O(\varepsilon^{-1} e^{-d_{\min}/\varepsilon})$ (a compact perturbation), it cannot change the sign of any eigenvalue that is bounded away from zero. By (a) and Weyl's perturbation theorem, the center eigenvalues converge to the Havelock values uniformly at $O(\varepsilon^\alpha)$. Therefore, for ε small enough that the center eigenvalues are within half their limiting values, the full Hessian inertia equals the center inertia—which is the point vortex Lagrangian inertia of Theorem 3. $\square$

Proposition 8 assumes a prescribed blob profile ϕ_ε . For a *general* $\mathbb{Z}_N$ -symmetric vorticity ω concentrated in N patches of radius ε near the N -gon vertices, the profile is not prescribed but determined by ω itself. The extension is immediate: decompose $\omega = \sum_k \omega_k$ via Voronoi cells, extract the center $\bar{z}_k = (1/\kappa) \int x \omega_k(x) dx$ and second moment $\sigma_k^2 \leq \varepsilon^2$ of each patch. The $\mathbb{Z}_N$ symmetry forces all patches to be congruent (up to rotation). The inter-patch Hessian depends on the patches' profiles only through their convolution with G —the same structure as Proposition 7—while the self-energy Hessian remains positive definite in H^{-1} for any profile. The non-symmetric shape corrections enter at $O(\varepsilon^3/d_{\min}^5)$ (the dipole moment vanishes by the center extraction; the first non-trivial correction is the quadrupole at order ε^2 relative to the monopole). Therefore Proposition 8 extends to arbitrary $\mathbb{Z}_N$ -symmetric vorticity patches: the stability boundary $N \leq 7$ holds for *any* smooth vorticity distribution sufficiently concentrated near the N -gon, not only for prescribed-profile blobs.

6.6.3 The Rossby bridge (Saturn regime)

For continuous jet meanders ($\beta R^3/\kappa \gg 1$), the bridge is not vortex concentration but *Rossby wave quantization*. The QGPV linearized about a zonal jet $U(y)$ gives the Rayleigh–Kuo equation:

$$(U - c)(\hat{\psi}'' - k^2\hat{\psi}) + (\beta - U'')\hat{\psi} = 0.$$

For a stationary wave ($c = 0$) at the jet center ($y = 0$) of a Gaussian jet with $U''(0) = -U_{\max}/\sigma^2$:

$$\hat{\psi}'' + \left[\frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2} \right] \hat{\psi} = 0, \quad \beta_{\text{eff}} = \beta + \frac{U_{\max}}{\sigma^2},$$

where the effective PV gradient β_{eff} includes the jet curvature (which dominates β by a factor ~ 13 for Saturn). The stationary azimuthal wavenumber is

$$n^* = R_S \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}, \quad (13)$$

where $R_S \cos \varphi$ is the ring radius at latitude φ ($\approx 1.31 \times 10^7$ m for Saturn at 76° N). The nearest integer $n = \text{round}(n^*)$ is the selected polygon wavenumber. For Saturn: $\beta_{\text{eff}}/U_{\max} \approx 1.72 \times 10^{-13} \text{ m}^{-2}$, giving $n^* \approx 5.4\text{--}5.6$ (the full WKB eigenfunction correction raises the value toward 6).

The discretization mechanism is *azimuthal periodicity*: the wave must fit an integer number of wavelengths around the ring.

The connection to the point vortex energy landscape proceeds through the *critical layer* of the saturated wave. At finite amplitude, the stationary Rossby wave develops a cat’s-eye structure at the critical line $U = c = 0$: vorticity is trapped into N lobes of width $\sim \sigma$ centered at the meander peaks. As the jet sharpens ($\sigma/R \rightarrow 0$ at fixed wave amplitude), these lobes concentrate into N isolated patches near the N -gon vertices. By the general vorticity patch result (following Proposition 8), the energy-Casimir Hessian of this N -patch configuration converges to the point vortex Hessian. The Saturn bridge thus arrives at the *same* energy landscape as the Jupiter bridge, through a different limiting procedure: Saturn concentrates a continuous jet meander into N patches (jet width $\sigma \rightarrow 0$), while Jupiter concentrates N discrete cyclones into point vortices (blob width $\varepsilon \rightarrow 0$).

6.6.4 The regime criterion

The dimensionless number $\beta R^3/\kappa$ is the ratio of two timescales: the Rossby wave period $T_R \sim 1/(\beta R)$ and the vortex turnover time $T_v \sim R^2/\kappa$. When $\beta R^3/\kappa = T_v/T_R \ll 1$, the vortex interactions complete many orbits before the Rossby wave propagates appreciably, so the dynamics is vortex-dominated. When $\beta R^3/\kappa \gg 1$, the Rossby wave dynamics operates faster than the vortex rearrangement, so the wave mechanism dominates:

Regime	$\beta R^3/\kappa$	Bridge
Vortex-dominated	$\ll 1$	Blob convergence (Proposition 7)
Wave-dominated	$\gg 1$	Rossby stationarity (equation (13))

Jupiter’s north pole ($\beta R^3/\kappa \approx 0.6$): discrete cyclones, Thomson stability applies, blob bridge proves convergence. Saturn’s hexagon ($\beta R^3/\kappa \approx 7.6$): continuous jet meander, Rossby stationarity selects $n = 6$, periodicity provides quantization. Both mechanisms require the 2D logarithmic Green’s function as the common mathematical kernel.

6.7 The Riemannian Havelock identity

On the sphere, the constrained Lagrangian eigenvalues at *every* tested colatitude ($\varphi_0 = 5^\circ$ through 90° , for $N = 6$) satisfy:

$$\lambda_m = C_1(\varphi_0) - \frac{m(N-m)}{2}, \quad (14)$$

with C_1 depending on the surface geometry but $f(m, N) = m(N-m)/2$ *universal*—the same on the sphere as on the plane (verified: $C_2 = 1.000$ at all colatitudes, to four decimal places; eigenvalue pair sums $\lambda_m^+ + \lambda_m^- = 2C_1$ constant across all modes at each colatitude).

The universality follows from the proof structure of Appendix A: the Havelock identity $T_m = m(N-m)/2$ is derived from (i) the algebraic factorization $|\sum_{k=0}^{m-1} \omega^{pk}|^2$ and (ii) the orthogonality of $\mathbb{Z}_N$ characters, both of which are properties of the discrete rotation group $\mathbb{Z}_N$ —not of the specific Green’s function. The Green’s function enters only through: $H_{rt}^{(m)} = 0$ (anti-symmetry of the sine sum, requiring only that G depends on geodesic distance) and $H_{tt}^{(m)} = -H_{rr}^{(m)}$ (tracelessness, from harmonicity of G away from sources). These two conditions hold for the Green’s function of the Laplace–Beltrami operator on *any* 2D Riemannian surface.

The stability threshold N_{crit} is therefore determined by $C_1(\text{surface}) \geq \max_m m(N-m)/2$: a condition involving the *universal* function f and a single surface-dependent constant. On the plane: $C_1 = N-1$, giving $N_{\text{crit}} = 7$ (the largest integer with $(N-1) \geq \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$). On the sphere: $C_1(\varphi_0) < N-1$ for $\varphi_0 > 0$, giving the colatitude-dependent threshold of Section 6.1.

The value $C_1 = N-1$ on the plane has a geometric origin: for $h' = -1/d$ (the logarithmic force law), each vortex pair $(0, k)$ contributes the *same* amount $-1/(2R)$ to the gradient $\partial H / \partial x_0$, regardless of the separation k . This is because $h'(d_k)(x_0 - x_k)/d_k = -(1 - \cos(2\pi k/N))/(4R \sin^2(\pi k/N)) = -1/(2R)$: the geometric factor $(1 - \cos)$ cancels against the denominator $\sin^2 = (1 - \cos)/2$. The cancellation is the *scale invariance* of the logarithmic interaction ($h' \propto 1/d$ compensates the d -dependence of the chord geometry), which is a consequence of *conformal invariance* in two dimensions. No other interaction has this equal-contribution property.

The number 7 is therefore the meeting point of two structures: *algebra* ($\mathbb{Z}_N$ orthogonality gives $f = m(N-m)/2$, universal across all surfaces) and *geometry* (2D conformal invariance gives $C_1 = N-1$, specific to the plane). The threshold $N_{\text{crit}} = 7$ is the largest integer satisfying $(N-1) \geq \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$ —a universal constant of 2D conformal geometry.

A The Havelock identity

Lemma 9. *For integers $N \geq 2$ and $1 \leq m \leq N-1$,*

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2}.$$

This identity appears in Havelock [1931], §3, as a step in computing the stability eigenvalues of the N -vortex ring; see also Aref et al. [2003], Appendix A.

Proof. Write $\omega = e^{2\pi i/N}$. Using $2 \sin(\pi p/N) = |1 - \omega^p|$ and the factorization $\sin(\pi pm/N)/\sin(\pi p/N) = \sum_{k=0}^{m-1} e^{i\pi p(m-1-2k)/N}$, we have

$$\frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{1}{2} \left| \sum_{k=0}^{m-1} \omega^{pk} \right|^2 - \frac{1}{2} \left(\sum_{k=0}^{m-1} \omega^{pk} \right)^2 \cdot (\text{phase}).$$

More directly: expand $[1 - \cos(2\pi pm/N)]/[4 \sin^2(\pi p/N)] = \frac{1}{2} \left| \sum_{k=0}^{m-1} \omega^{pk} \right|^2 = \frac{1}{2} \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \omega^{p(j-k)}$. Summing over $p = 1, \dots, N-1$ and using the orthogonality relation $\sum_{p=1}^{N-1} \omega^{p\ell} = N\delta_{\ell \equiv 0 \pmod{N}} - 1$:

$$T_m = \frac{1}{2} \sum_{j,k=0}^{m-1} (N\delta_{j \equiv k \pmod{N}} - 1) = \frac{1}{2} (Nm - m^2) = \frac{m(N-m)}{2},$$

where $\delta_{j \equiv k}$ contributes m (since $0 \leq j, k \leq m-1 < N$ forces $j = k$) and the -1 contributes $-m^2$. $\square$

References

- A. Adriani, A. Mura, G. Orton, et al. Clusters of cyclones encircling Jupiter’s poles. *Nature*, 555:216–219, 2018.
- Alex B. Akins, Mark D. Hofstadter, Bryan J. Butler, A. James Friedson, Edward M. Molter, Robert J. Sault, and Imke de Pater. Evidence of a polar cyclone on Uranus from VLA observations. *Geophysical Research Letters*, 50:e2023GL102872, 2023.
- H. Aref, P. K. Newton, M. A. Stremler, T. Tokieda, and D. L. Vainchtein. Vortex crystals. *Advances in Applied Mechanics*, 39:1–79, 2003.
- A. C. Barbosa Aguiar, P. L. Read, R. D. Wordsworth, T. Salter, and Y. Hiro Yamazaki. A laboratory model of Saturn’s north polar hexagon. *Icarus*, 206:755–763, 2010.
- S. Boatto and H. E. Cabral. Nonlinear stability of a latitudinal ring of point-vortices on a nonrotating sphere. *SIAM Journal on Applied Mathematics*, 64:216–230, 2003.
- Freddy Bouchet and Antoine Venaille. Statistical mechanics of two-dimensional and geophysical flows. *Physics Reports*, 515:227–295, 2012.
- P. H. Chavanis and J. Sommeria. Classification of self-organized vortices in two-dimensional turbulence: the case of a bounded domain. *Journal of Fluid Mechanics*, 314:267–297, 1996.
- Nimrod Gavriel and Yohai Kaspi. The number and location of Jupiter’s circumpolar cyclones explained by vorticity dynamics. *Nature Geoscience*, 14:559–563, 2021.
- D. A. Godfrey. A hexagonal feature around Saturn’s north pole. *Icarus*, 76:335–356, 1988.
- T. H. Havelock. The stability of motion of rectilinear vortices in ring formation. *Philosophical Magazine*, 11:617–633, 1931.

- Thomas R. N. Jansson, Martin P. Haspang, Kare H. Jensen, Pascal Hersen, and Tomas Bohr. Polygons on a rotating fluid surface. *Physical Review Letters*, 96:174502, 2006.
- Robert H. Kraichnan. Inertial ranges in two-dimensional turbulence. *Physics of Fluids*, 10:1417–1423, 1967.
- R. Morales-Juberías, K. M. Sayanagi, T. E. Dowling, and A. A. Simon-Miller. Simulations of the structure of Saturn’s hexagonal jet. *Icarus*, 211:1284–1293, 2011.
- Lars Onsager. Statistical hydrodynamics. *Il Nuovo Cimento*, 6:279–287, 1949.
- J. Pedlosky. *Geophysical Fluid Dynamics*. Springer-Verlag, 2nd edition, 1987.
- A. Sánchez-Lavega, T. del Río-Gaztelurrutia, R. Hueso, et al. The long-term steady motion of Saturn’s hexagon and the stability of its enclosed jet stream under seasonal changes. *Geophysical Research Letters*, 41:1425–1431, 2014.
- Lia Siegelman, William R. Young, and Andrew P. Ingersoll. Polar vortex crystals: Emergence and structure. *Proceedings of the National Academy of Sciences*, 119(17):e2120486119, 2022.
- J. J. Thomson. A treatise on the motion of vortex rings. *Macmillan, London*, 1883.
- G. K. Vallis. *Atmospheric and Oceanic Fluid Dynamics*. Cambridge University Press, 2nd edition, 2017.